

Zhaoxiang (Aaron) Feng

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Education

The University of Texas at Austin

Sep. 2026 – Present

M.S. in Computer Science

University of California San Diego

Sep. 2022 – Mar. 2026

B.S. in Data Science & B.S. in Probability & Statistics

- GPA 3.84 (Major GPAs: 3.95 Data Science, 3.93 Statistics); Provost Honors.
- Graduate coursework: Machine Learning, Time Series Analysis, Applied Statistics, Optimization, Privacy-sensitive Data Systems.

Publications

[1] Matthew Ho, Chen Si, **Zhaoxiang Feng**, Fangxu Yu, Yichi Yang, Zhijian Liu, Zhiting Hu, Lianhui Qin. “ArcMemo: Abstract Reasoning Composition with Lifelong LLM Memory.”

Runner Up, ARC Prize 2025 Paper Awards, Top 8 of 90 submissions.

[\[paper\]](#) [\[code\]](#)

[2] Lanxiang Hu, **Zhaoxiang Feng**, Yulun Wu, Haoran Yuan, Yujie Zhao, Yu-Yang Qian, Bojun Wang, Peng Zhao, Daxin Jiang, Yibo Zhu, Tajana Rosing, Hao Zhang. “JetSpec: Breaking the Scaling Ceiling of Speculative Decoding with Parallel Tree Drafting.”

Under review at Conference on Neural Information Processing Systems (NeurIPS), 2026.

[\[paper\]](#) [\[code\]](#)

[3] **Zhaoxiang Feng**, Mingyang Yao, Charlie Sun, David Scott Lewis. “Which Memory Operation Drives Recovery? A Factorial Study of Retrieve, Write, and Manage Adaptation under Domain Shift.”

Accepted at ICLR 2026 Workshop on Lifelong Agents: Learning, Aligning, Evolving (LLA), 2026.

[\[paper\]](#)

Research Experience

Research Assistant | Hao AI Lab, UC San Diego

Sep. 2025 – Present

Advisors: Prof. Hao Zhang and Lanxiang Hu

- Designed **JetSpec++**, a causal correction for parallel drafting that conditions each drafted token on its full branch history rather than the immediately preceding token. Achieves 0.89 additional accepted tokens per verification round over a parameter-matched baseline, with the margin widening at draft depths beyond those used in training.
- Contributed to **JetSpec**, a draft head that generates a complete speculative tree in one forward pass while preserving the target model’s autoregressive factorization along every branch; built its serving-engine internals and evaluation suite. Delivers up to 9.64× end-to-end speedup on Qwen3-8B.
- Ported **DFlash** speculative decoding to TPU in JAX, exploiting the hardware’s tile structure to remove a GPU-motivated cap on draft width. Averaged 3.13× speedup on TPU v5p; merged into **vLLM’s TPU backend** and featured on the [Google Developers Blog](#).
- Built the distributed training stack for a 1.8B-parameter language model on eight NVIDIA B200 GPUs, implementing pipeline parallelism and conducting scaling analysis.

Research Assistant | Q-Lab, UC San Diego

Jan. 2025 – Jun. 2026

Advisors: Prof. Lianhui Qin and Matthew Ho

- Led the design of **ArcMemo**’s program-synthesis memory ontology: a typed, parameterized representation in which reusable reasoning concepts are stored and composed.
- Replaced embedding-based retrieval with reasoning-guided concept selection, yielding a 7.5% relative gain over the no-memory baseline on ARC-AGI-1 (59.33% official score) and 15% over embedding retrieval.
- Built the concept dataset pipeline, converting hand-written concepts into validated training puzzles through staged LLM generation, code synthesis, and automated testing.
- Extended the framework to AIME competition mathematics with a metacognitive gate governing when memory is consulted, improving accuracy by 9.3%; conducted the memory-format comparisons that settled

the final design.

Student Researcher | Halicioğlu Data Science Institute, UC San Diego

Sep. 2025 – Present

Advisor: Prof. Yian Ma

- Developing **Triad**, a diagnostic framework for interactive reasoning agents that localizes whether task performance is capped by model capability, harness design, or available context, decomposing success into question targeting, evidence acquisition, and answer conversion.
- On AR-Bench detective cases, a ladder of structural interventions over the framework's slot-structured knowledge graph raised accuracy on the pinned 50-case hard set from 18% to 46%; the same structure ported to the MediQ clinical benchmark without domain-specific modification.
- Established evidence acquisition, rather than answer accuracy, as the framework's progress signal: substituting a weaker adversary raises accuracy from 27% to 40% while acquisition falls from 24% to 16%.

Student Researcher | Wang Lab, UC San Diego

Jun. 2025 – Jun. 2026

Advisors: Prof. Wei Wang and Young Su Ko

- Developed **TrustPPI**, a model-agnostic reliability signal for protein–protein interaction prediction that perturbs a model's internal representations and measures prediction stability. Manuscript in preparation.
- Achieved 0.66–0.83 AUROC for error detection across three architectures; stability-guided selection confirms true interactions 2–3.5× faster than confidence-based baselines.

Teaching & Service

Tutor & Grader

Fall 2024 – Fall 2025

University of California San Diego

- **Tutor:** set theory, probability, and algorithmic thinking, Summer 2025.
- **Grader:** probability, stochastic processes, mathematical statistics, and computational statistics, 7 quarters.

Technical Skills

Languages: Python, R, SQL, Java, JavaScript

ML frameworks: PyTorch, JAX, Hugging Face Transformers

Systems & infrastructure: LLM serving engines (vLLM, SGLang), distributed GPU training, TPU inference (JAX), Docker, AWS